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Electronic device having ceramic element and metal terminals

Abstract

An electronic device includes: a ceramic element including a first electrode portion and a second electrode portion; a first metal terminal including a first external connection portion and a first electrode connection portion connected to the first electrode portion through solder; and a second metal terminal including a second external connection portion and a second electrode connection portion connected to the second electrode portion through solder. The first electrode connection portion includes a first solder stopper formed by undulation of at least a part of a facing surface of the first electrode connection portion facing the first electrode portion to control a range of an adhesion region of solder to the first electrode portion. The second electrode connection portion includes a second solder stopper that is formed by undulation of at least a part of a facing surface of the second electrode connection portion facing the second electrode portion.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present disclosure relates to an electronic device used as a capacitor and the like.

Description of the Related Art

(2) Conventionally, an electronic device such as a capacitor formed by connecting metal terminals to a single-plate dielectric disk as disclosed in JP H6-196348 A has been proposed.

(3) However, in the conventional electronic device, especially when a high voltage is applied, connection failures may occur between the dielectric disk and the metal terminals.

CITATION LIST

(4) Patent Document 1: JP H6-196348 A

BRIEF SUMMARY OF THE INVENTION

(5) The present disclosure has been made in view of such circumstances, and it is an object of the present disclosure to provide an electronic device with improved pressure resistance.

(6) In order to achieve the aforementioned object, an electronic device according to the present

disclosure includes: a ceramic element including a first electrode portion and a second electrode portion; a first metal terminal including a first electrode connection portion and a first external connection portion, the first electrode connection portion being connected to the first electrode portion through solder; and a second metal terminal including a second electrode connection portion and a second external connection portion, the second electrode connection portion being connected to the second electrode portion through solder. The first electrode connection portion includes a first solder stopper formed by undulation of at least a part of a facing surface of the first electrode connection portion facing the first electrode portion to control a range of an adhesion region of solder to the first electrode portion, and the second electrode connection portion includes a second solder stopper formed by undulation of at least a part of a facing surface of the second electrode connection portion facing the second electrode portion to control a range of an adhesion region of solder to the second electrode portion.

(7) The present inventors have found that as the amount of solder for connecting the electrode connection portion of the metal terminal and the electrode portion of the ceramic element to each other increases, the pressure resistance decreases. The pressure resistance of the electronic device can be improved by controlling the adhesion region of the solder for connecting the electrode connection portion and the electrode portion to be within a predetermined range by using the solder stopper of the electrode connection portion.

(8) At least one of the first solder stopper and the second solder stopper includes an electrode proximity portion on the facing surface and a solder storage portion for storing solder. In addition, the solder storage portion includes a recessed portion extending in a direction away from the ceramic element. Furthermore, an edge of the recessed portion is closed by the electrode proximity portion. By storing the solder in the recessed portion with the closed edge, the solder adhesion region can be controlled. As a result, the pressure resistance of the electronic device can be improved.

(9) At least one of the first solder stopper and the second solder stopper includes a solder outflow prevention portion for controlling spread of solder, which is formed adjacent to an electrode proximity portion on the facing surface. In addition, the solder outflow prevention portion includes a recessed portion extending in a direction away from the ceramic element. Furthermore, a part of an edge of the recessed portion has an opening portion that is not closed by the electrode proximity portion. Since solder stays in the recessed portion with the open edge, the outflow of the solder can be effectively prevented. Therefore, the solder adhesion region can be controlled.

(10) The first electrode connection portion has an approximately rectangular shape extending along a first main surface of the ceramic element. The second electrode connection portion has an approximately rectangular shape extending along a second main surface of the ceramic element. At least one of the first electrode connection portion and the second electrode connection portion has the solder outflow prevention portion arranged so as to extend in a direction crossing a longitudinal direction of the first electrode connection portion or the second electrode connection portion. In addition, the adhesion region and a non-adhesion region of the first electrode portion or the second electrode portion to which solder does not adhere are separated from each other with the solder outflow prevention portion as a boundary.

(11) By adopting such a configuration, the solder adhesion region can be controlled to be located on one side in the longitudinal direction of the electrode connection portion. As a result, the pressure resistance of the electronic device can be improved.

(12) At least one of the first solder stopper and the second solder stopper includes a protruding portion that protrudes in a direction approaching the ceramic element. The outflow of the solder can also be prevented by forming a protruding portion in the electrode connection portion.

(13) A case in which the ceramic element is housed in a housing recess may be provided. Since the ceramic element is housed in the housing recess of the case, there is no need for a step of arranging the ceramic element and the like in a cavity for resin molding and performing molding with an

exterior material. As a result, good productivity is obtained.

(14) The first external connection portion and the second external connection portion are arranged in an opening edge portion of the housing recess. By adopting such a configuration, when the electronic device is mounted on a substrate or the like, a connection to an external circuit formed on the substrate can be easily made with the opening edge portion as a mounting surface.

(15) In addition, the first external connection portion and the second external connection portion are arranged approximately parallel to a first main surface and a second main surface of the ceramic element. In addition, the first external connection portion and the second external connection portion are arranged approximately perpendicular to a first main surface and a second main surface of the ceramic element.

(16) the housing recess is filled with mold resin. By adopting such a configuration, the mold resin can cover the entire ceramic element. As a result, the strength, insulation, and the like of the electronic device can be improved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic perspective view showing the configuration of an electronic device according to a first embodiment;

(2) FIG. 2 is a schematic perspective view of the electronic device shown in FIG. 1 when viewed from another angle;

(3) FIG. 3 is a schematic perspective view showing the configuration of a case of the electronic device shown in FIG. 1;

(4) FIG. 4 is a schematic perspective view showing the configuration of a ceramic element of the electronic device shown in FIG. 1;

(5) FIG. 5 is a schematic perspective view showing the configuration of a metal terminal of the electronic device shown in FIG. 1;

(6) FIG. 6 is a schematic perspective view showing a partially assembled state of the electronic device shown in FIG. 1;

(7) FIG. 7 is a plan view of a part of the electronic device shown in FIG. 6;

(8) FIG. 8 is a cross-sectional view taken along the line VIII-VIII shown in FIG. 1;

(9) FIG. 9 is a schematic perspective view showing the configuration of a metal terminal of an electronic device according to a second embodiment;

(10) FIG. 10 is a plan view showing a partially assembled state of the electronic device according to the second embodiment;

(11) FIG. 11A is a cross-sectional view of the electronic device according to the second embodiment;

(12) FIG. 11B is a cross-sectional view showing a modification example of the electronic device according to the second embodiment;

(13) FIG. 12 is a schematic perspective view showing the configuration of an electronic device according to a third embodiment;

(14) FIG. 13 is a schematic perspective view of the electronic device shown in FIG. 12 when viewed from another angle;

(15) FIG. 14 is a schematic perspective view showing the configuration of a case of the electronic device shown in FIG. 12;

(16) FIG. 15 is a schematic perspective view showing the configuration of a metal terminal of the electronic device shown in FIG. 12;

(17) FIG. 16 is a plan view showing a partially assembled state of the electronic device according to the third embodiment;

- (18) FIG. 17 is a cross-sectional view taken along the line XVII-XVII shown in FIG. 12;
(19) FIG. 18 is a schematic perspective view showing the configuration of a metal terminal of an electronic device according to a fourth embodiment;
(20) FIG. 19 is a plan view showing a partially assembled state of the electronic device according to the fourth embodiment; and
(21) FIG. 20 is a cross-sectional view of the electronic device according to the fourth embodiment.

DETAILED DESCRIPTION OF THE INVENTIONS

(22) Hereinafter, the present disclosure will be described based on embodiments shown in the diagrams. For those who provide explanations with reference to diagrams as necessary, the illustrated contents are only schematically and exemplarily shown for understanding of the present application, and the appearance, dimensional ratio, and the like may be different from the actual ones. In addition, the present disclosure will be specifically described below based on embodiments, but the present disclosure is not limited to these embodiments.

First Embodiment

- (23) FIG. 1 is a schematic perspective view of an electronic device 10 according to the present embodiment. As shown in FIG. 1, the electronic device 10 includes a ceramic element 30 and a case 20 in which the ceramic element 30 is housed. As shown in FIG. 2, the case 20 includes a housing recess 21, and the ceramic element 30 is housed in the housing recess 21.
- (24) As shown in FIG. 4, the ceramic element 30 has the first main surface 31a and the second main surface 31b facing each other, and has an approximately disk-like outer shape. However, the ceramic element 30 may have a shape other than the disk shape, such as an elliptical disk shape or a rectangular flat plate shape. The first main surface 31a and the second main surface 31b are a pair of surfaces having the largest area in the ceramic element 30. In addition, as shown in FIG. 8, in the description of the ceramic element 30, one of the two main surfaces is referred to as the first main surface 31a, and the other main surface is referred to as the second main surface 31b. However, the first main surface 31a and the second main surface 31b may be interchanged with respect to the state shown in the diagram.
- (25) As shown in FIG. 8, which is a cross-sectional view of the electronic device 10, the ceramic element 30 includes a first electrode portion 33 formed on the first main surface 31a, a second electrode portion 34 formed on the second main surface 31b, and a dielectric portion 35 interposed between the first electrode portion 33 and the second electrode portion 34. The material of the dielectric portion 35 is not particularly limited. For example, the dielectric portion 35 is formed of a dielectric material such as calcium titanate, strontium titanate, barium titanate, or a mixture thereof. In addition, the ceramic element 30 is not limited to a capacitor or the like in which the dielectric portion 35 is interposed between the first electrode portion 33 and the second electrode portion 34. For example, the ceramic element may be a varistor or a thermistor in which a semiconductor ceramic is interposed between a first electrode portion and a second electrode portion.
- (26) The materials of the first electrode portion 33 and the second electrode portion 34 are not particularly limited, and copper, a copper alloy, nickel, a nickel alloy, and the like are usually used. However, silver, an alloy of silver and palladium, and the like can also be used. The thicknesses of the first electrode portion 33 and the second electrode portion 34 are usually about 10 to 50 μm although the thicknesses are not particularly limited. In addition, a metal coating film containing at least one selected from Ni, Cu, Sn, and the like may be formed on the surfaces of the first electrode portion 33 and the second electrode portion 34.
- (27) As shown in FIGS. 1 and 2, the case 20 defines the outer shape of the electronic device 10, and the outer shape of the electronic device 10 is an approximately hexagonal prism shape. All outer side surfaces 24, 25, 26a, 26b, 28a, and 28b of the case 20 are not parallel to the first main surface 31a and the second main surface 31b of the ceramic element 30. However, the electronic device 10 is not limited thereto, and may have a polygonal prism shape other than the hexagonal prism shape, or may have a flat plate shape such as a rectangular parallelepiped.

(28) As shown in FIG. 3, the case **20** has six outer side surfaces perpendicular to an opening edge portion **22** surrounding an opening **21a** of the housing recess **21**. The six outer side surfaces of the case **20** are a first outer side surface **24**, a second outer side surface **25**, a third outer side surface **26b**, a fourth outer side surface **28a**, a fifth outer side surface **26a**, and a sixth outer side surface **28b**.

(29) As shown in FIG. 8, the first outer side surface **24** and the second outer side surface **25** are a pair of outer side surfaces facing each other. As shown in FIG. 8, the fifth outer side surface **26a** is connected to the first outer side surface **24** while forming an included angle greater than 90 degrees, and the third outer side surface **26b** is connected to the second outer side surface **25** while forming an included angle greater than 90 degrees. The fifth outer side surface **26a** and the third outer side surface **26b** have approximately the same area, and are connected to each other while forming an included angle greater than 90 degrees at the central portion of the electronic device **10**.

(30) As shown in FIG. 8, the fourth outer side surface **28a** and the sixth outer side surface **28b** are symmetrical with respect to the fifth outer side surface **26a** and the third outer side surface **26b**, respectively. That is, the fourth outer side surface **28a** is connected to the first outer side surface **24** while forming an included angle greater than 90 degrees, and the sixth outer side surface **28b** is connected to the second outer side surface **25** while forming an included angle greater than 90 degrees. The fourth outer side surface **28a** and the sixth outer side surface **28b** have approximately the same area, and are connected to each other while forming an included angle greater than 90 degrees at the central portion of the electronic device **10**.

(31) As shown in FIG. 8, the first and second outer side surfaces **24** and **25** are approximately perpendicular to the first main surface **31a** and the second main surface **31b**, whereas the fourth to sixth outer side surfaces are arranged obliquely with respect to the first main surface **31a** and the second main surface **31b**.

(32) As shown in FIGS. 2 and 3, the case **20** includes the housing recess **21** and the opening edge portion **22** surrounding the opening **21a** of the housing recess **21**. In the electronic device **10**, the side where the opening **21a** of the housing recess **21** formed in the case **20** is formed is a mounting surface side facing a substrate that is a mounting target when mounting the electronic device **10**.

(33) FIG. 2 is a bottom view of the electronic device **10** when viewed from the mounting surface side. In the case **20**, the ceramic element **30** and the like are housed inside the housing recess **21**. The opening **21a** of the housing recess **21** has a shape smaller than the outer shape of the case **20**. In addition, as shown in FIG. 1, a top surface **23** of the case **20** is formed above the electronic device **10** in a direction opposite to the opening **21a** of the housing recess **21**.

(34) As shown in FIG. 8, the housing recess **21** has six inner side surfaces along the outer shape of the case **20**. A first inner side surface **241** is parallel to the first outer side surface **24**, a second inner side surface **251** is parallel to the second outer side surface **25**, a third inner side surface **26b1** is parallel to the third outer side surface **26b**, a fourth inner side surface **28a1** is parallel to the fourth outer side surface **28a**, a fifth inner side surface **26a1** is parallel to the fifth outer side surface **26a**, and a sixth inner side surface **28b1** is parallel to the sixth outer side surface **28b**.

(35) As shown in FIG. 8, a support portion **291** that protrudes vertically toward the first main surface **31a1** of the ceramic element **30** and a support portion **292** that protrudes vertically toward the second main surface **31b** are formed in the housing recess **21**. The support portion **291** is arranged between the third inner side surface **26b1** and the fifth inner side surface **26a1**, and the support portion **292** is arranged between the fourth inner side surface **28a1** and the sixth inner side surface **28b1**. As shown in FIG. 3, the support portions **291** and **292** extend from the opening edge portion **22** to a bottom surface **21b** of the case **20**. In addition, although the support portions **291** and **292** may not be provided, the support portions **291** and **292** can stabilize the ceramic element housed in the housing recess **21**.

(36) As shown in FIGS. 2 and 3, the opening edge portion **22** is configured by a plane surrounding the opening **21a**. The opening edge portion **22** is approximately perpendicular to the first main

surface **31a** and the second main surface **31b** of the ceramic element **30**. In the present embodiment, the opening edge portion **22** is even with the ends of the support portions **291** and **292**. In addition, the shape of the opening edge portion **22** may change depending on the shape of the opening **21a**, the outer peripheral shape of the case **20**, and the like.

(37) As shown in FIG. 2, the ceramic element **30** is arranged in the housing recess **21** so that the first main surface **31a** and the second main surface **31b** face a direction perpendicular to the opening **21a**. Therefore, as shown in FIG. 3, a depth **W2**, which is the distance from the opening edge portion **22** to the bottom surface **21b** of the case **20**, is larger than a diameter **W1** of the ceramic element **30** shown in FIG. 4. Therefore, in the electronic device **10**, the entire ceramic element **30** can be housed in the housing recess **21** without a part of the ceramic element **30** being exposed from the opening **21a** of the housing recess **21**. In addition, in the present embodiment, as shown in FIG. 8, in the electronic device **10**, the housing recess **21** is filled with mold resin **70**, and the entire ceramic element **30** can be covered with the mold resin **70**.

(38) FIG. 5 is a schematic perspective view showing a first metal terminal **40** and a second metal terminal **50** in the present embodiment. As shown in FIG. 1, the electronic device **10** includes a pair of first and second metal terminals **40** and **50**. The first metal terminal **40** and the second metal terminal **50** are arranged so as to be spaced apart from each other in the electronic device **10**, thereby being electrically insulated from each other. The first metal terminal **40** and the second metal terminal **50** are formed, for example, by machining a conductive metal plate material. However, the method of forming the first and second metal terminals **40** and **50** is not particularly limited.

(39) As shown in FIGS. 6 and 7, the first metal terminal **40** includes a first electrode connection portion **44** connected to the first electrode portion **33** of the ceramic element **30** through solder, a first external connection portion **42** arranged in the opening edge portion **22** shown in FIGS. 1 and 2, a first terminal arm portion **46** that connects the first external connection portion **42** and the first electrode connection portion **44** to each other, and a first folded portion **47** extending upward from a distal end **42a** of the first external connection portion **42**. As shown in FIG. 2, the first electrode connection portion **44** and the first terminal arm portion **46** are housed in the housing recess **21** of the case **20** similarly to the ceramic element **30**.

(40) As shown in FIG. 7, the first electrode connection portion **44** has an approximately rectangular shape extending approximately parallel along the first main surface **31a** of the ceramic element **30**. As shown in FIG. 8, a facing surface **44a** of the first electrode connection portion **44** is connected to the first electrode portion **33** formed on the first main surface **31a** through solder **60**. On the other hand, the first external connection portion **42** is arranged on one side of the opening edge portion **22** (on the first outer side surface **24** side of the case **20**).

(41) As shown in FIG. 1, the first terminal arm portion **46** of the first metal terminal **40** connects the first electrode connection portion **44** inside the housing recess of the case **20** to the first external connection portion **42** outside the housing recess of the case **20**. In addition, as shown in FIG. 1, the first external connection portion **42** is approximately perpendicular to the first main surface **31a** and the second main surface **31b** of the ceramic element **30**.

(42) As shown in FIG. 1, the distal end **42a** of the first external connection portion **42** is an end on a side opposite to a side of the first external connection portion **42** connected to the first terminal arm portion **46**, and is connected to the first folded portion **47**.

(43) As shown in FIG. 1, the first folded portion **47** of the first metal terminal **40** extends upward along the first outer side surface **24** of the case **20** from the distal end **42a** of the first external connection portion **42**. Since the first metal terminal **40** includes the first folded portion **47**, a solder fillet is easily formed during mounting. Therefore, the electronic device **10** including such a first folded portion **47** has good mountability. In addition, as shown in FIG. 1, since the first terminal arm portion **46**, the first external connection portion **42**, and the first folded portion **47** are provided, the distal end of the first metal terminal **40** has a J shape. For this reason, in the electronic

device **10**, a part of the case **20** is interposed between the first terminal arm portion **46** and the first folded portion **47**. Therefore, the first metal terminal **40** can be more strongly engaged with the case **20**.

(44) As shown in FIG. **8**, the first electrode connection portion **44** includes a first solder stopper **440**. The first solder stopper **440** is formed by the undulation of a part of the facing surface **44a** of the first electrode connection portion **44** facing the first electrode portion **33**. In the first solder stopper **440**, the range of an adhesion region **61** of the solder **60** to the first electrode portion **33** is controlled.

(45) As shown in FIG. **5**, in the present embodiment, the first solder stopper **440** includes an electrode proximity portion **44a1** on the facing surface **44a**. As shown in FIG. **8**, the first solder stopper **440** includes a solder storage portion **441** for storing the solder **60**. The electrode proximity portion **44a1** is a portion of the facing surface **44a** closest to the first main surface **31a** of the first electrode portion **33**. In the present embodiment, the electrode proximity portion **44a1** includes an edge **442a** of a recessed portion **442**, and corresponds to a glue margin for bonding the first metal terminal **40** and the ceramic element **30** to each other with the solder **60** stored in the solder storage portion **441**. In addition, the electrode proximity portion **44a1** may be in contact with the first electrode portion **33**. The solder storage portion **441** includes the recessed portion **442** extending in a direction away from the ceramic element **30** due to the undulation of the facing surface **44a** in the electrode proximity portion **44a1**. That is, the recessed portion **442** is recessed with respect to the ceramic element **30**.

(46) The edge **442a** of the recessed portion **442** shown in FIG. **8** is closed by the electrode proximity portion **44a1** as shown in FIG. **5**. The solder **60** adheres to the electrode proximity portion **44a1**, so that the first metal terminal **40** and the first electrode portion **33** are electrically connected to each other. By storing the solder **60** inside the recessed portion **442** with the edge **442a** closed, the solder adhesion region **61** can be controlled.

(47) As shown in FIG. **5**, the second metal terminal **50** has the same shape and size as the first metal terminal **40**. As shown in FIG. **6**, the second metal terminal **50** includes a second electrode connection portion **54** connected to the second electrode portion **34** of the ceramic element **30** through solder, a second external connection portion **52** arranged in the opening edge portion **22** shown in FIGS. **1** and **2**, a second terminal arm portion **56** that connects the second external connection portion **52** and the second electrode connection portion **54** to each other, and a second folded portion **57** extending upward from a distal end **52a** of the second external connection portion **52**. As shown in FIG. **2**, the second electrode connection portion **54** and the second terminal arm portion **56** are housed in the housing recess **21** of the case **20** similarly to the ceramic element **30**.

(48) As shown in FIG. **7**, the second electrode connection portion **54** has an approximately rectangular shape extending approximately parallel along the second main surface **31b** of the ceramic element **30**. As shown in FIG. **8**, a facing surface **54a** of the second electrode connection portion **54** is connected to the second electrode portion **34** formed on the second main surface **31b** through solder. On the other hand, the second external connection portion **52** is arranged on one side of the opening edge portion **22** (on the second outer side surface **25** side of the case **20**).

(49) As shown in FIG. **1**, the second terminal arm portion **56** of the second metal terminal **50** connects the second electrode connection portion **54** inside the housing recess of the case **20** to the second external connection portion **52** outside the housing recess of the case **20**. In addition, as shown in FIG. **1**, the second external connection portion **52** is approximately perpendicular to the first main surface **31a** and the second main surface **31b** of the ceramic element **30**.

(50) As shown in FIG. **2**, the distal end **52a** of the second external connection portion **52** is an end on a side opposite to a side of the second external connection portion **52** connected to the second terminal arm portion **56**, and is connected to the second folded portion **57**.

(51) As shown in FIG. **1**, the second folded portion **57** of the second metal terminal **50** extends

upward along the second outer side surface **25** of the case **20** from the distal end **52a** of the second external connection portion **52**. Since the second metal terminal **50** includes the second folded portion **57**, a solder fillet is easily formed during mounting. Therefore, the electronic device **10** including such a second folded portion **57** has good mountability. In addition, as shown in FIG. **1**, since the second terminal arm portion **56**, the second external connection portion **52**, and the second folded portion **57** are provided, the distal end of the second metal terminal **50** has a J shape. For this reason, in the electronic device **10**, a part of the case **20** is interposed between the second terminal arm portion **56** and the second folded portion **57**. Therefore, the second metal terminal **50** can be more strongly engaged with the case **20**.

(52) As shown in FIG. **8**, the second electrode connection portion **54** includes a second solder stopper **540**. The second solder stopper **540** is formed by the undulation of a part of the facing surface **54a** of the second electrode connection portion **54** facing the second electrode portion **34**. In the second solder stopper **540**, the range of an adhesion region **62** of the solder **60** to the second electrode portion **34** is controlled.

(53) As shown in FIG. **5**, in the present embodiment, the second solder stopper **540** includes the electrode proximity portion **54a1** on the facing surface **54a**. As shown in FIG. **8**, the second solder stopper **540** includes a solder storage portion **541** for storing the solder **60**. The electrode proximity portion **54a1** is a portion of the facing surface **54a** closest to the second main surface **31b** of the second electrode portion **34**. In the present embodiment, the electrode proximity portion **54a1** includes an edge **542a** of a recessed portion **542**, and corresponds to a glue margin for bonding the second metal terminal **50** and the ceramic element **30** to each other with the solder **60** stored in the solder storage portion **541**. In addition, the electrode proximity portion **54a1** may be in contact with the second electrode portion **34**. The solder storage portion **541** includes the recessed portion **542** extending in a direction away from the ceramic element **30** due to the undulation of the facing surface **54a** in the electrode proximity portion **54a1**. That is, the recessed portion **542** is recessed with respect to the ceramic element **30**.

(54) The edge **542a** of the recessed portion **542** shown in FIG. **8** is closed by the electrode proximity portion **54a1** as shown in FIG. **5**. The solder **60** adheres to the electrode proximity portion **54a1**, so that the second metal terminal **50** and the second electrode portion **34** are electrically connected to each other. By storing the solder **60** inside the recessed portion **542** with the edge **542a** closed, the solder adhesion region **62** can be controlled.

(55) Normally, a ceramic element in an electronic device repeats expansion and contraction due to piezoelectric action or the like especially when a high AC voltage is applied to the ceramic element, but metal terminals have little change in volume. In addition, solder that connects the ceramic element and the metal terminals constrains the expansion and contraction of the ceramic element. The present inventors have found that force is applied to the solder due to the constraint of the expansion and contraction of the ceramic element, causing cracks or the like.

(56) As shown in FIG. **8**, in the present embodiment, the edges **442a** and **542a** of the recessed portions **442** and **542** of the solder storage portions **441** and **541** are closed, and the solder **60** is stored therein. By controlling the areas of the solder adhesion regions **61** and **62** to be small in this manner, the ceramic element **30** is not excessively constrained by the first electrode connection portion **44** and the second electrode connection portion **54**. Therefore, in the electronic device **10** according to the present embodiment, the pressure resistance is improved. For example, even at medium and high voltages of 8 kV to 20 kV, the electronic device **10** can be used satisfactorily without causing cracks or the like in the ceramic element **30**. In addition, in this specification, solder is a term that represents not only an alloy containing lead and tin as main components but also a conductive adhesive and a material that enables electrical connection and fixing between members.

(57) In addition, since the solder **60** is stored in the solder storage portions **441** and **541**, deviations in the amount of solder applied, deviations in the areas of the solder adhesion regions **61** and **62**,

and deviation in the degree of adhesion between the ceramic element **30** and the first electrode connection portion **44** and the second electrode connection portion **54** can also be reduced. The areas of the solder adhesion regions **61** and **62** are not particularly limited as long as these can secure the electrical connection between the first electrode portion **33** and the first electrode connection portion **44** and the electrical connection between the second electrode portion **34** and the second electrode connection portion **54**. For example, the areas of the solder adhesion regions **61** and **62** can be one fourth or less of the areas of the facing surfaces **44a** and **54a**. In addition, as shown in FIG. 5, in the present embodiment, the recessed portions **442** and **542** are circular with the edges **442a** and **542a** closed, but are not limited to this shape, and may be elliptical, polygonal, or the like.

(58) The materials of the first metal terminal **40** and the second metal terminal **50** shown in FIG. 5 and the like are not particularly limited as long as these are metal materials having electrical conductivity. For example, iron, nickel, copper, silver, or alloys containing these can be used. In addition, a metal coating film containing Ni, Sn, Cu, or the like may be formed on the surfaces of the first metal terminal **40** and the second metal terminal **50**.

(59) As shown in FIG. 2, in the present embodiment, the first external connection portion **42** and the second external connection portion **52** are arranged in the opening edge portion **22** of the housing recess **21**, but the present disclosure is not limited thereto. In this configuration, when the electronic device **10** is mounted on a substrate or the like, a connection to an external circuit formed on the substrate can be easily made with the opening edge portion **22** as a mounting surface.

(60) As shown in FIG. 1, in the present embodiment, the electronic device **10** includes the case **20**, but the present disclosure is not limited thereto. When there is the case **20** in which the ceramic element **30** is housed in the housing recess **21**, there is no need for a step of arranging the ceramic element **30** and the like in a cavity for resin molding and performing molding with an exterior material because the ceramic element **30** is housed in the housing recess **21** of the case **20**. Therefore, good productivity is obtained.

(61) As shown in FIG. 8, in the present embodiment, the housing recess **21** is filled with a mold resin **70**. In this configuration, since the mold resin **70** can cover the entire ceramic element **30**, the strength, insulation, and the like of the electronic device **10** can be improved. However, the housing recess **21** may not be filled with resin, and a gap may be formed between the inner wall of the housing recess **21** and the ceramic element **30** and each of the first and second metal terminals **40** and **50**.

(62) The case **20** can be manufactured, for example, by injection molding using resin. However, the material of the case **20** is not limited to resin.

(63) The electronic device **10** shown in FIG. 1 can be manufactured, for example, by the following steps. First, the ceramic element **30**, the first metal terminal **40**, and the second metal terminal **50** are prepared, molten solder is stored in the solder storage portions **441** and **541** of the first and second metal terminals **40** and **50**, and the first metal terminal **40** and the second metal terminal **50** are connected to the first electrode portion **33** and the second electrode portion **34** of the ceramic element **30**, respectively. Alternatively, in a state in which molten solder is placed at predetermined positions of the first electrode portion **33** and the second electrode portion **34**, the connections may be made so that the molten solder is placed in the solder storage portions **442** and **542** of the first and second electrode connection portions **44** and **54**.

(64) Then, an intermediate product shown in FIG. 6 in which the first metal terminal **40**, the second metal terminal **50**, and the ceramic element **30** are integrated is arranged in the housing recess **21** of the case **20**. Thereafter, if necessary, mold resin is injected into the housing recess **21** shown in FIG. 2 to obtain the electronic device **10** shown in FIG. 1. As described above, in the electronic device **10** shown in FIG. 2, the ceramic element **30** is housed in the housing recess **21** of the case **20**. Therefore, since there is no need for a step of arranging the ceramic element **30** and the like in a

cavity for resin molding and performing molding with an exterior material, good productivity is obtained.

(65) In the electronic device **10**, the ceramic element **30** and the like can be housed in the case **20**. Therefore, the size of the ceramic element **30** can be flexibly changed as long as the ceramic element **30** can be housed in the case **20**. In addition, since the first external connection portion **42** and the second external connection portion **52** are perpendicular to the first main surface **31a** and the second main surface **31b** of the ceramic element **30**, the mounting area of the electronic device **10** can be reduced, which is suitable for surface mounting.

Second Embodiment

(66) An electronic device according to the present embodiment is different from the electronic device **10** according to the first embodiment in terms of the first solder stopper **440** and the second solder stopper **540**, but the basic configuration is the same as that in the first embodiment. In the present embodiment, the description of portions in common with the first embodiment will be omitted, and different portions will mainly be described in detail below. Portions not described in the present embodiment are the same as those described in the first embodiment.

(67) FIG. **9** is a schematic perspective view of the first metal terminal **40** and the second metal terminal **50** of the electronic device **10** according to the present embodiment. FIG. **10** is a plan view of the assembly of the ceramic element **30**, the first metal terminal **40**, and the second metal terminal **50** of the electronic device **10** according to the present embodiment. FIG. **11A** is a cross-sectional view of the electronic device **10** according to the present embodiment.

(68) As shown in FIG. **11A**, in the present embodiment, the first solder stopper **440** is formed adjacent to the electrode proximity portion **44a1** on the facing surface **44a** of the first electrode connection portion **44**. In the present embodiment, the electrode proximity portion **44a1** corresponds to a portion with which the solder **60** is in contact. As shown in FIG. **9**, in the present embodiment, the solder outflow prevention portion **443** is arranged so as to extend in a direction crossing the longitudinal direction of the first electrode connection portion **44**.

(69) As shown in FIG. **11A**, in the present embodiment, the first solder stopper **440** includes the solder outflow prevention portion **443** for controlling the spread of the solder **60**. The solder outflow prevention portion **443** includes a recessed portion **444** extending in a direction away from the first main surface **31a** of the ceramic element **30**. That is, the recessed portion **442** is recessed with respect to the ceramic element **30**. As shown in FIG. **9**, at an edge **444a** of the recessed portion **444**, an opening portion **444a1** that is not closed by the electrode proximity portion **44a1** is formed. At an edge **444b** of the recessed portion **444**, an opening portion **444b1** that is not closed by the electrode proximity portion **44a1** is formed.

(70) As shown in FIG. **11A**, in the present embodiment, the second solder stopper **540** is formed adjacent to the electrode proximity portion **54a1** on the facing surface **54a** of the second electrode connection portion **54**. In the present embodiment, the electrode proximity portion **54a1** corresponds to a portion with which the solder **60** is in contact. As shown in FIG. **9**, in the present embodiment, a solder outflow prevention portion **543** is arranged so as to extend in a direction crossing the longitudinal direction of the second electrode connection portion **54**.

(71) As shown in FIG. **11A**, in the present embodiment, the second solder stopper **540** includes the solder outflow prevention portion **543** for controlling the spread of the solder **60**. The solder outflow prevention portion **543** includes a recessed portion **544** extending in a direction away from the second main surface **31b** of the ceramic element **30**. That is, the recessed portion **542** is recessed with respect to the ceramic element **30**. As shown in FIG. **9**, at an edge **544a** of the recessed portion **544**, an opening portion **544a1** that is not closed by the electrode proximity portion **54a1** is formed. At an edge **544b** of the recessed portion **544**, an opening portion **544b1** that is not closed by the electrode proximity portion **54a1** is formed.

(72) As shown in FIG. **11A**, in the present embodiment, the solder **60** adheres to the facing surfaces **44a** and **54a** between the solder outflow prevention portions **443** and **543** and ends **48** and **58**. In

addition, the solder **60** adheres to the first electrode portion **33** and the second electrode portion **34** to form the adhesion regions **61** and **62**. The adhesion regions **61** and **62** and non-adhesion regions **63** and **64** to which the solder **60** does not adhere in the first electrode portion **33** or the second electrode portion **34** are separated from each other with the solder outflow prevention portions **443** and **543** as boundaries.

(73) In the present embodiment, on the facing surface **44a** of the first electrode connection portion **44** of the first metal terminal **40**, molten solder is placed between the end **48** and the solder outflow prevention portion **443**, and the first electrode connection portion **44** is pressed against the first main surface **31a** of the ceramic element **30** so that the molten solder spreads. As a result, the first metal terminal **40** and the ceramic element **30** can be electrically connected to each other in the solder adhesion region **61**.

(74) In the recessed portion **444** of the solder outflow prevention portion **443**, no pressing force acts on the molten solder spread between the first electrode portion **33** and the facing surface **44a**. Therefore, the spread of the molten solder spread between the first electrode portion **33** and the facing surface **44a** in the direction of the first terminal arm portion **46** is controlled. In this manner, the solder adhesion region **61** can be controlled to be located on the end **48** side of the first electrode connection portion **44**.

(75) Similarly to the first metal terminal **40**, on the facing surface **54a** of the second electrode connection portion **54** of the second metal terminal **50**, molten solder is placed between the end **58** and the solder outflow prevention portion **543**, and the second electrode connection portion **54** is pressed against the second main surface **31b** of the ceramic element **30** so that the molten solder spreads. As a result, the second metal terminal **50** and the ceramic element **30** can be electrically connected to each other in the solder adhesion region **62**.

(76) In the recessed portion **544** of the solder outflow prevention portion **543**, no pressing force acts on the molten solder spread between the second electrode portion **34** and the facing surface **54a**. Therefore, the spread of the molten solder spread between the second electrode portion **34** and the facing surface **54a** in the direction of the second terminal arm portion **56** is controlled. In this manner, the solder adhesion region **62** can be controlled to be located on the end **58** side of the second electrode connection portion **54**.

(77) Thus, in the present embodiment, the areas of the solder adhesion regions **61** and **62** are controlled to be small. For this reason, the ceramic element **30** is not excessively constrained by the first electrode connection portion **44** and the second electrode connection portion **54**. Therefore, in the electronic device **10** according to the present embodiment, the pressure resistance can be improved.

(78) In the present embodiment, the first solder stopper **440** and the second solder stopper **540** include the recessed portions **444** and **544** extending in a direction away from the ceramic element **30**, respectively, but the present disclosure is not limited thereto. For example, as shown in FIG. **11B**, the first solder stopper **440** and the second solder stopper **540** may include protruding portions **45** and **55** that protrude in a direction approaching the ceramic element **30**, respectively.

(79) By adopting such a configuration, the spread of the solder **60** can be controlled by the distal end portions of the protruding portions **45** and **55** formed in the first electrode connection portion **44** and the second electrode connection portion **54** and the surfaces of the protruding portions **45** and **55** on the ends **48** and **58** side. In this case, the distal end portions of the protruding portions **45** and **55** become the electrode proximity portions **44a1** and **54a1**, respectively, and are connected to the solder **60**. Moreover, between the first electrode portion **33** and the facing surface **44a**, the solder adhesion regions **61** and **62** are arranged closer to the ends **48** and **58** than the protruding portions **45** and **55**. Therefore, the spread of the solder in the direction of the first terminal arm portion **46** and the second terminal arm portion **56** is controlled.

Third Embodiment

(80) An electronic device according to the present embodiment is different from the electronic

device **10** according to the first embodiment in terms of the first metal terminal and the second metal terminal **50**, but the basic configuration is the same as that in the first embodiment. In the present embodiment, the description of portions in common with the first embodiment will be omitted, and different portions will mainly be described in detail below. Portions not described in the present embodiment are the same as those described in the first embodiment.

(81) As shown in FIGS. **12** and **13**, the external shape of the electronic device **10** according to the present embodiment is an approximately rectangular parallelepiped flat plate shape. The ceramic element **30** has the first main surface **31a** and the second main surface **31b** arranged approximately parallel to the top surface **23**. As shown in FIG. **13**, the case **20** of the present embodiment has four outer side surfaces perpendicular to the opening edge portion **22** surrounding the opening **21a** of the housing recess **21**. The four outer side surfaces of the case **20** are a first outer side surface **24**, a second outer side surface **25**, a third outer side surface **260**, and a fourth outer side surface **280**.

(82) As shown in FIG. **14**, the first outer side surface **24** and the second outer side surface **25** are a pair of outer side surfaces facing each other. In addition, the third outer side surface **260** and the fourth outer side surface **280** are a pair of outer side surfaces facing each other. The first outer side surface **24** and the second outer side surface **25** are perpendicularly connected to the third outer side surface **260** and the fourth outer side surface **280**.

(83) As shown in FIG. **12**, the first outer side surface **24**, the second outer side surface **25**, the third outer side surface **260**, and the fourth outer side surface **280** are arranged approximately perpendicular to the first main surface **31a** and the second main surface **31b**.

(84) As shown in FIG. **14**, the housing recess **21** has four inner side surfaces along the outer shape of the case **20**. The first inner side surface **241** is parallel to the first outer side surface **24**, the second inner side surface **251** is parallel to the second outer side surface **25**, the third inner side surface **261** is parallel to the third outer side surface **260**, and the fourth inner side surface **281** is parallel to the fourth outer side surface **280**.

(85) As shown in FIG. **14**, a solder stopper arrangement hole **290** is formed on the bottom surface **21b** of the housing recess **21**. As shown in FIG. **17**, a bottom portion **290a** of the solder stopper arrangement hole **290** is deeper than the bottom surface **21b**. Therefore, the first solder stopper **440** is not in contact with the case **20** when the first solder stopper **440** is arranged in the solder stopper arrangement hole **290**.

(86) As shown in FIG. **13**, the ceramic element **30** is arranged in the housing recess **21** so that the second main surface **31b** faces the same direction as the opening **21a**, without a part of the ceramic element **30** being exposed.

(87) FIG. **15** is a schematic perspective view showing the first metal terminal **40** and the second metal terminal **50** in the present embodiment. FIG. **16** is a plan view of the assembly of the ceramic element **30**, the first metal terminal **40**, and the second metal terminal **50** of the electronic device **10** according to the present embodiment. FIG. **17** is a cross-sectional view of the electronic device **10** according to the present embodiment.

(88) As shown in FIG. **17**, in the first metal terminal **40**, the facing surface **44a** of the first electrode connection portion **44** is connected to the first electrode portion **33** formed on the first main surface **31a** through solder. On the other hand, the first external connection portion **42** is arranged on the first outer side surface **24** side of the opening edge portion **22**. In addition, the first external connection portion **42** is approximately parallel to the first main surface **31a** and the second main surface **31b** of the ceramic element **30**.

(89) As shown in FIG. **17**, in the present embodiment, the first terminal arm portion **460** of the first metal terminal **40** has one end connected to the first electrode connection portion **44** and the other end extending to the second main surface **31b** side of the ceramic element **30**. The other end of the first terminal arm portion **460** is connected to the first external connection portion **42**.

(90) As shown in FIG. **15**, the distal end **42a** of the first external connection portion **42** is an end on a side opposite to a side of the first external connection portion **42** connected to the first terminal

arm portion **46**, and is connected to the first folded portion **47**.

(91) As shown in FIG. **17**, in the second metal terminal **50**, the facing surface **54a** of the second electrode connection portion **54** is connected to the second electrode portion **34** formed on the second main surface **31b** through solder. On the other hand, the second external connection portion **52** is arranged on the first outer side surface **25** side of the opening edge portion **22**. In addition, the second external connection portion **52** is approximately parallel to the first main surface **31a** and the second main surface **31b** of the ceramic element **30**.

(92) As shown in FIG. **17**, in the present embodiment, the second terminal arm portion **560** of the second metal terminal **50** has one end connected to the second electrode connection portion **54** and the other end extending to the second main surface **31b** side of the ceramic element **30**. The other end of the second terminal arm portion **560** is connected to the second external connection portion **52**.

(93) As shown in FIG. **15**, the distal end **52a** of the second external connection portion **52** is an end on a side opposite to a side of the second external connection portion **52** connected to the second terminal arm portion **56**, and is connected to the second folded portion **57**.

(94) As shown in FIG. **13**, in the electronic device **10** according to the present embodiment, the first external connection portion **42** and the second external connection portion **52** are arranged in the opening edge portion **22** of the case **20**, and the opening **21a** side of the housing recess **21** is a mounting surface side facing a substrate that is a mounting target. In addition, in the present embodiment, the second main surface **31b** of the ceramic element **30** faces the same direction as the opening **21a** of the housing recess **21**, so that the height of the electronic device **10** is reduced. In addition, since the area of the electronic device **10** on the mounting surface side is large, the electronic device **10** is more stable when mounted.

(95) In the present embodiment, the first solder stopper **440** and the second solder stopper **540** have the same configuration as the first solder stopper **440** and the second solder stopper **540** according to the first embodiment, and accordingly, have the same effect.

(96) That is, as shown in FIG. **17**, the first solder stopper **440** includes the solder storage portion **441** in which the solder **60** can be stored. The solder storage portion **441** includes the recessed portion **442** extending in a direction away from the ceramic element **30** due to the undulation of the facing surface **44a** in the electrode proximity portion **44a1**.

(97) In addition, the second solder stopper **540** includes the solder storage portion **541** in which the solder **60** can be stored. The solder storage portion **541** includes the recessed portion **542** extending in a direction away from the ceramic element **30** due to the undulation of the facing surface **54a** in the electrode proximity portion **54a1**.

(98) With such a configuration, in the present embodiment, the areas of the solder adhesion regions **61** and **62** are controlled to be small. Therefore, the pressure resistance can be improved.

Fourth Embodiment

(99) An electronic device according to the present embodiment is different from the electronic device **10** according to the third embodiment in terms of the first solder stopper **440** and the second solder stopper **540**, but the basic configuration is the same as that in the third embodiment. In the present embodiment, the description of portions in common with the third embodiment will be omitted, and different portions will mainly be described in detail below. Portions not described in the present embodiment are the same as those described in the third embodiment.

(100) FIG. **18** is a schematic perspective view of the first metal terminal **40** and the second metal terminal **50** according to the present embodiment. FIG. **19** is a plan view of the assembly of the ceramic element **30**, the first metal terminal **40**, and the second metal terminal **50** of the electronic device **10** according to the present embodiment. FIG. **20** is a cross-sectional view of the electronic device **10** according to the present embodiment.

(101) In the present embodiment, the first solder stopper **440** and the second solder stopper **540** have the same configuration as the first solder stopper **440** and the second solder stopper **540**

according to the second embodiment, and accordingly, have the same effect.

(102) That is, as shown in FIG. 20, the first solder stopper **440** includes the solder outflow prevention portion **443** for controlling the spread of the solder **60**. The solder outflow prevention portion **443** includes the recessed portion **444** extending in a direction away from the first main surface **31a** of the ceramic element **30**. The solder outflow prevention portion **443** is arranged so as to extend in a direction crossing the longitudinal direction of the first electrode connection portion **44**.

(103) In addition, the second solder stopper **540** includes the solder outflow prevention portion **543** for controlling the spread of the solder **60**. The solder outflow prevention portion **543** includes the recessed portion **544** extending in a direction away from the first main surface **31a** of the ceramic element **30**. The solder outflow prevention portion **543** is arranged so as to extend in a direction crossing the longitudinal direction of the second electrode connection portion **54**. With such a configuration, in the present embodiment, the areas of the solder adhesion regions **61** and **62** are controlled to be small. Therefore, the pressure resistance can be improved.

(104) In addition, the above-described embodiments include, within the scope of the technical scope, forms with various design changes that do not depart from the gist of the claims.

(105) For example, a solder storage portion or a solder outflow prevention portion that forms a solder stopper may be formed so that the facing surface of an electrode connection portion is undulated so as to prevent the solder adhesion region from spreading excessively, or a plurality of solder storage portions or solder outflow prevention portions may be provided in one electrode connection portion.

EXPLANATIONS OF LETTERS OR NUMERALS

(106) **10** ELECTRONIC DEVICE **20** CASE **21** HOUSING RECESS **21a** OPENING **21b** BOTTOM SURFACE **22** OPENING EDGE PORTION **23** TOP SURFACE **24** FIRST OUTER SIDE SURFACE **25** SECOND OUTER SIDE SURFACE **26a** FIFTH OUTER SIDE SURFACE **26b**, **260** THIRD OUTER SIDE SURFACE **28a**, **280** FOURTH OUTER SIDE SURFACE **28b** SIXTH OUTER SIDE SURFACE **241** FIRST INNER SIDE SURFACE **251** SECOND INNER SIDE SURFACE **26a1** FIFTH INNER SIDE SURFACE **26b1**, **261** THIRD INNER SIDE SURFACE **28a1**, **281** FOURTH INNER SIDE SURFACE **28b1** SIXTH INNER SIDE SURFACE **291**, **292** SUPPORT PORTION **290** SOLDER STOPPER ARRANGEMENT HOLE **290a** BOTTOM PORTION **30** CERAMIC ELEMENT **31a** FIRST MAIN SURFACE **33** FIRST ELECTRODE PORTION **31b** SECOND MAIN SURFACE **34** SECOND ELECTRODE PORTION **35** DIELECTRIC PORTION **40** FIRST METAL TERMINAL **42** FIRST EXTERNAL CONNECTION PORTION **42a** DISTAL END **44** FIRST ELECTRODE CONNECTION PORTION **44a** FACING SURFACE **44a1** ELECTRODE PROXIMITY PORTION **440** FIRST SOLDER STOPPER **441** SOLDER STORAGE PORTION **442** RECESSED PORTION **442a** EDGE **443** SOLDER OUTFLOW PREVENTION PORTION **444** RECESSED PORTION **444a**, **444b** EDGE **444a1**, **445b1** OPENING PORTION **45** PROTRUDING PORTION **48** END **47** FIRST FOLDED PORTION **46**, **460** FIRST TERMINAL ARM PORTION **50** SECOND METAL TERMINAL **52** SECOND EXTERNAL CONNECTION PORTION **52a** DISTAL END **54** SECOND ELECTRODE CONNECTION PORTION **54a** FACING SURFACE **54a1** ELECTRODE PROXIMITY PORTION **540** SECOND SOLDER STOPPER **541** SOLDER STORAGE PORTION **542** RECESSED PORTION **542a** EDGE **543** SOLDER OUTFLOW PREVENTION PORTION **544** RECESSED PORTION **544a**, **544b** EDGE **544a1**, **545b1** OPENING PORTION **55** PROTRUDING PORTION **58** END **57** SECOND FOLDED PORTION **56**, **560** SECOND TERMINAL ARM PORTION **60** SOLDER **61**, **62** ADHESION REGION **63**, **64** NON-ADHESION REGION **70** MOLD RESIN

Claims

1. An electronic device, comprising: a ceramic element including a first electrode portion and a second electrode portion; a first metal terminal including a first electrode connection portion and a first external connection portion, the first electrode connection portion being connected to the first electrode portion through solder; and a second metal terminal including a second electrode connection portion and a second external connection portion, the second electrode connection portion being connected to the second electrode portion through solder, wherein the first electrode connection portion includes a first solder stopper formed by undulation of at least a part of a facing surface of the first electrode connection portion facing the first electrode portion to control a range of an adhesion region of solder to the first electrode portion, the second electrode connection portion includes a second solder stopper formed by undulation of at least a part of a facing surface of the second electrode connection portion facing the second electrode portion to control a range of an adhesion region of solder to the second electrode portion, at least one of the first solder stopper and the second solder stopper includes an electrode proximity portion on the facing surface and a solder storage portion for storing solder, the solder storage portion includes a recessed portion extending in a direction away from the ceramic element, and an edge of the recessed portion is closed by the electrode proximity portion.
2. The electronic device according to claim 1, wherein at least one of the first solder stopper and the second solder stopper includes a solder outflow prevention portion for controlling spread of solder, which is formed adjacent to an electrode proximity portion on the facing surface.
3. The electronic device according to claim 2, wherein the solder outflow prevention portion includes a recessed portion extending in a direction away from the ceramic element.
4. The electronic device according to claim 2, wherein the solder outflow prevention portion includes a recessed portion extending in a direction away from the ceramic element, and a part of an edge of the recessed portion has an opening portion that is not closed by the electrode proximity portion.
5. The electronic device according to claim 1, further comprising a case in which the ceramic element is housed in a housing recess.
6. The electronic device according to claim 5, wherein the first external connection portion and the second external connection portion are arranged on an opening edge portion of the housing recess.
7. The electronic device according to claim 6, wherein the first external connection portion and the second external connection portion are arranged approximately parallel to a first main surface and a second main surface of the ceramic element.
8. The electronic device according to claim 6, wherein the first external connection portion and the second external connection portion are arranged approximately perpendicular to a first main surface and a second main surface of the ceramic element.
9. An electronic device, comprising: a ceramic element including a first electrode portion and a second electrode portion; a first metal terminal including a first electrode connection portion and a first external connection portion, the first electrode connection portion being connected to the first electrode portion through solder; and a second metal terminal including a second electrode connection portion and a second external connection portion, the second electrode connection portion being connected to the second electrode portion through solder, wherein the first electrode connection portion includes a first solder stopper formed by undulation of at least a part of a facing surface of the first electrode connection portion facing the first electrode portion to control a range of an adhesion region of solder to the first electrode portion, the second electrode connection portion includes a second solder stopper formed by undulation of at least a part of a facing surface of the second electrode connection portion facing the second electrode portion to control a range of an adhesion region of solder to the second electrode portion, at least one of the first solder stopper and the second solder stopper includes a solder outflow prevention portion for controlling spread of solder, which is formed adjacent to an electrode proximity portion on the facing surface, the first

electrode connection portion has an approximately rectangular shape extending along a first main surface of the ceramic element, the second electrode connection portion has an approximately rectangular shape extending along a second main surface of the ceramic element, and at least one of the first electrode connection portion and the second electrode connection portion has the solder outflow prevention portion arranged so as to extend in a direction crossing a longitudinal direction of the first electrode connection portion or the second electrode connection portion.

10. The electronic device according to claim 9, wherein the adhesion region and a non-adhesion region of the first electrode portion or the second electrode portion to which solder does not adhere are separated from each other with the solder outflow prevention portion as a boundary.

11. The electronic device according to claim 9, wherein at least one of the first solder stopper and the second solder stopper includes a protruding portion that protrudes in a direction approaching the ceramic element.

12. The electronic device according to claim 9, wherein at least one of the first solder stopper and the second solder stopper includes an electrode proximity portion on the facing surface and a solder storage portion for storing solder.

13. The electronic device according to claim 12, wherein the solder storage portion includes a recessed portion extending in a direction away from the ceramic element.

14. An electronic device, comprising: a ceramic element including a first electrode portion and a second electrode portion; a case in which the ceramic element is housed in a housing recess; a first metal terminal including a first electrode connection portion and a first external connection portion, the first electrode connection portion being connected to the first electrode portion through solder; and a second metal terminal including a second electrode connection portion and a second external connection portion, the second electrode connection portion being connected to the second electrode portion through solder, wherein the first electrode connection portion includes a first solder stopper formed by undulation of at least a part of a facing surface of the first electrode connection portion facing the first electrode portion to control a range of an adhesion region of solder to the first electrode portion, the second electrode connection portion includes a second solder stopper formed by undulation of at least a part of a facing surface of the second electrode connection portion facing the second electrode portion to control a range of an adhesion region of solder to the second electrode portion, and the housing recess is filled with mold resin.
